Ex.No: 21 16/07/26

Basic Networking commands

Aim :-

Basic Networking commands in linux and windows operating systems.

Basic Networking commands :-

* **arp -a:**
  ARP is short form of address resolution protocol.

* **hostname:**
  This is the simplest of all TCP/IP commands.

* **ipconfig /all:**
  This commands displays detailed configuration information about connection including Router, Gateway, DNS, DHCP and type of Ethernet.

* **nbstat -a:**
  This command helps solve problems with Net BIOS name resolution for Net BIOS over TCP/IP.

* **net stat:**
  Net stat displays a variety of statistics about a computer TCP/IP connections.

```
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address
  192.168.56.255        ff-ff-ff-ff-ff-ff
  224.0.0.2             01-00-5e-00-00-02
  224.0.0.22            01-00-5e-00-00-16
  224.0.0.251           01-00-5e-00-00-fb
  224.0.0.252           01-00-5e-00-00-fc
  224.0.0.253           01-00-5e-00-00-fd
  224.0.1.60            01-00-5e-00-01-3c
  239.0.0.8             01-00-5e-00-00-08
  239.192.152.143       01-00-5e-40-98-8f
  239.255.67.250        01-00-5e-7f-43-fa
  239.255.102.18        01-00-5e-7f-66-12
  239.255.255.250       01-00-5e-7f-ff-fa
```

```
ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [MET

  -f            Clears the routing tables of all
                used in conjunction with one of
                cleared prior to running the com

  -p            When used with the ADD command,
                boots of the system. By default,
                when the system is restarted. Ig
                which always affect the appropri

  -4            Force using IPv4.

  -6            Force using IPv6.

PS C:\Users\DELL> hostname
HARISH-PC
PS C:\Users\DELL> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc4
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

(*) nslookup :-

It is a tool used to perform DNS lookups in display DNS details.

(*) pathping :-

pathping is unique to window's, and is basically a combination of Dracert commands.

(*) Ping :

Ping command is the best way to test connectivity nodes.

(*) Route :-

Route commands is used to show / manipulate the IP routing table.

Linux Networking commands :-

(*) IP :-

The ip command is one of the basic commands every administrator will need in daily work. The ip command can show address information, manipulate routing, plus display network various, interfaces and tunnels

(*) if config :

The if config command is a staple in many sysadmin's tool belt for configuring and troubleshooting.

```
PS C:\Users\DELL> netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            HDC0124118:0           LISTEN
  TCP    0.0.0.0:445            HDC0124118:0           LISTEN
  TCP    0.0.0.0:3306           HDC0124118:0           LISTEN
  TCP    0.0.0.0:5040           HDC0124118:0           LISTEN
  TCP    0.0.0.0:33060          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49664          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49665          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49666          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49667          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49668          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49676          HDC0124118:0           LISTEN
  TCP    0.0.0.0:49677          HDC0124118:0           LISTEN
  TCP    127.0.0.1:27017        HDC0124118:0           LISTEN
  TCP    127.0.0.1:49671        HDC0124118:49672       ESTABL
  TCP    127.0.0.1:49672        HDC0124118:49671       ESTABL
  TCP    127.0.0.1:49673        HDC0124118:49674       ESTABL
  TCP    127.0.0.1:49674        HDC0124118:49673       ESTABL

Users\DELL>pathping

ge: pathping [-g host-list] [-h maximum_hops]
             [-p period] [-q num_queries] [-w
             [-4] [-6] target_name

ions:
 -g host-list     Loose source route along ho
 -h maximum_hops  Maximum number of hops to s
 -i address       Use the specified source ad
 -n               Do not resolve addresses to
 -p period        Wait period milliseconds be
 -q num_queries   Number of queries per hop.
 -w timeout       Wait timeout milliseconds f
 -4               Force using IPv4.
 -6               Force using IPv6.
```

```
My traceroute  [v0.95]                    2026-07-16T1
st (::1)                                  Order of fields
ode  Restart statistics                          Pings
                  Packets
                Loss%  Snt  Last  Avg  Best
                0.0%   190   0.0   0.1   0.0
```

```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.25
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x2
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64
global>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 77  bytes 29443 (28.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 109  bytes 12886 (12.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2130 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2130 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

(*) mtr :-

MTR is a program with a command-line interface that services as a network diagnostic and troubleshooting tool.

(*) tcp dump :-

The tcp dump command is designed for capturing and displaying packets.

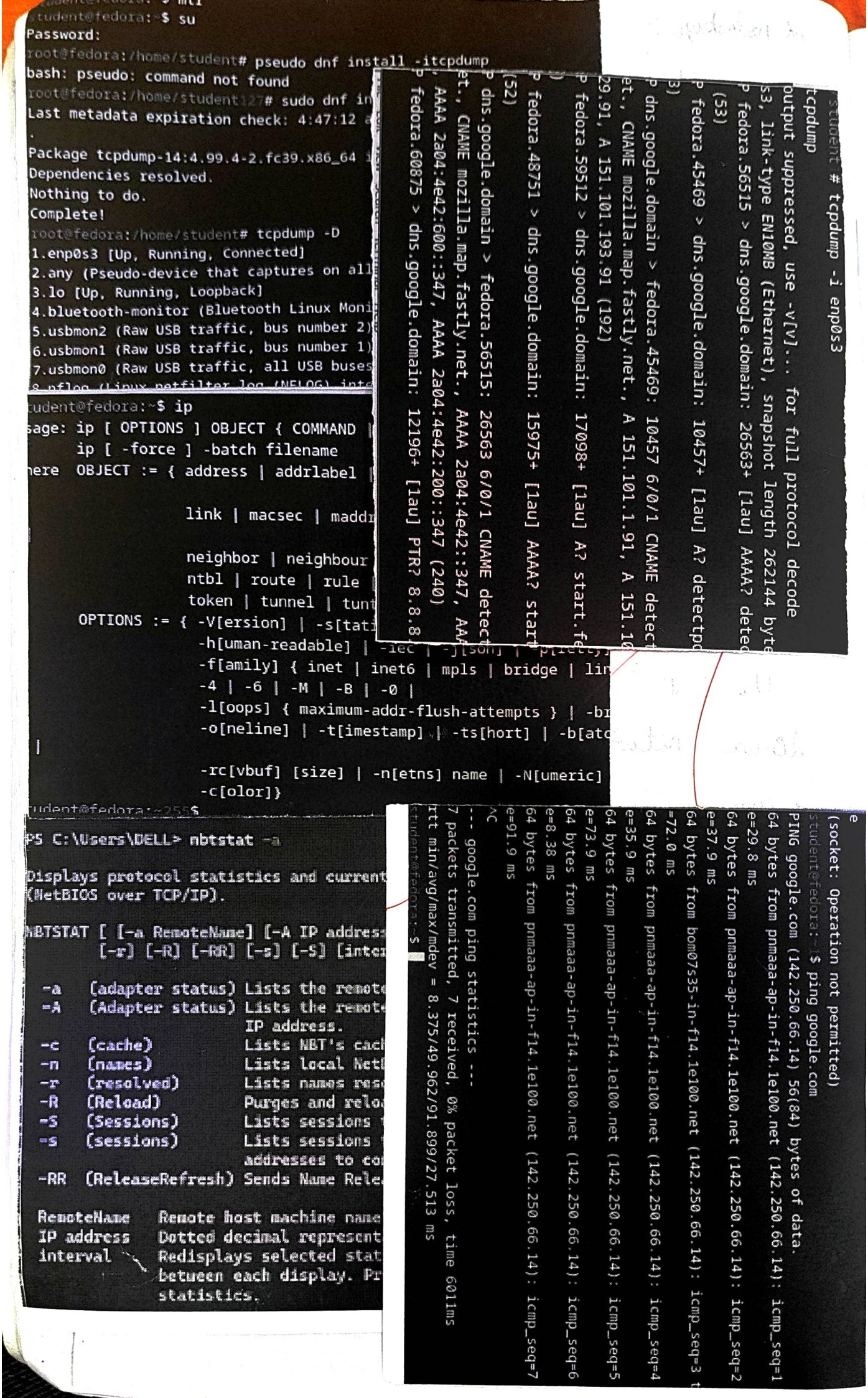
(*) ping :-

ping is a tool that verifies IP-level connectivity to another ICPI IP computer by sending Internet control message Protocol (ICMP)

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Result :-

Basic Networking commands in both Linux and windows operating systems is successfully executed and verified.


su
Password:
root@fedora:/home/student# pseudo dnf install -itcpdump
bash: pseudo: command not found
Last metadata expiration check: 4:47:12
Package tcpdump-14:4.99.4-2.fc39.x86_64
Dependencies resolved.
Nothing to do.
Complete!
root@fedora:/home/student# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Moni
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses
ip
Usage: ip [OPTIONS] OBJECT { COMMAND |
ip [-force] -batch filename
where OBJECT := { address | addrlabel |
link | macsec | maddr
neighbor | neighbour
ntbl | route | rule
token | tunnel | tunt
OPTIONS := { -V[ersion] | -s[tati
-h[uman-readable] | -iec | -j[son] | -p[ret
-f[amily] { inet | inet6 | mpls | bridge | lin
-4 | -6 | -M | -B | -0 |
-l[oops] { maximum-addr-flush-attempts } | -br
-o[neline] | -t[imestamp] | -ts[hort] | -b[atc
-rc[vbuf] [size] | -n[etns] name | -N[umeric]
-c[olor]}
tcpdump -i enp0s3
tcpdump
output suppressed, use -v[v]... for full protocol decode
s3, link-type EN10MB (Ethernet), snapshot length 262144 byte
fedora.56515 > dns.google.domain: 26563+ [1au] AAAA? detec
(53)
fedora.45469 > dns.google.domain: 10457+ [1au] A? detectpo
3)
dns.google.domain > fedora.45469: 10457 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., A 151.101.1.91, A 151.10
29.91, A 151.101.193.91 (192)
fedora.59512 > dns.google.domain: 17098+ [1au] A? start.fe
fedora.48751 > dns.google.domain: 15975+ [1au] AAAA? start
(52)
dns.google.domain > fedora.56515: 26563 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., AAAA 2a04:4e42::347, AA
AAAA 2a04:4e42:600::347, AAAA 2a04:4e42:200::347 (240)
fedora.60875 > dns.google.domain: 12196+ [1au] PTR? 8.8.8
PS C:\Users\DELL> nbtstat -a
Displays protocol statistics and current
(NetBIOS over TCP/IP).
NBTSTAT [[-a RemoteName] [-A IP address
[-r] [-R] [-RR] [-s] [-S] [inte
-a (adapter status) Lists the remote
-A (Adapter status) Lists the remote
IP address.
-c (cache) Lists NBT's cach
-n (names) Lists local NetB
-r (resolved) Lists names reso
-R (Reload) Purges and relo
-S (Sessions) Lists sessions
-s (sessions) Lists sessions
addresses to co
-RR (ReleaseRefresh) Sends Name Rele
RemoteName Remote host machine name
IP address Dotted decimal represent
interval Redisplays selected stat
betueen each display. Pr
statistics.
e
(socket: Operation not permitted)
student@fedora:~$ ping google.com
PING google.com (142.250.66.14) 56(84) bytes of data.
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1
e=29.8 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2
e=37.9 ms
64 bytes from bom07s35-in-f14.1e100.net (142.250.66.14): icmp_seq=3 t
=72.0 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4
e=35.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5
e=73.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6
e=8.38 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7
e=91.9 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 8.375/49.962/91.899/27.513 ms
student@fedora:~$